Old courses (left) to communities as new courses (right), Louvain split to <= 6 est. ECTS
flow width = credits of the old course spent on topics of the community; est. ECTS = sum of the community's topic ECTS (median per-course share)

B0009 Advanced Software Modelling and Design (6 CFU)
B8018 DIGITAL HEALTH M (8 CFU)
81676 Digital Forensic (6 CFU)
93363 Advanced Programming (6 CFU)
28106 Computer Science for Environmental Sciences (6 CFU)
07276 Computer Science (3 CFU)
B4826 Informatics for the Humanities (3 CFU)
96346-B Computer Science (2 CFU)
57828 Informatics (3 CFU)
78510 Informatics (2 CFU)
70210 Lab-based Course on Computer Graphics (2 CFU)
07276-C Computer Science (2 CFU)
B8396 Basic IT Skills (1 CFU)
B9254 Basic IT Skills (1 CFU)
15920 Computer Lab (1 CFU)
07276-B Computer Science (1 CFU)
B0011 Software Process Engineering (6 CFU)
95627 Software Engineering (6 CFU)
B0010 Software Architecture and Platforms (6 CFU)
B2133 Big Data Analytics and Text Mining (6 CFU)
40720 Data Mining (6 CFU)
95631-A Machine Learning and Data Mining (6 CFU)
91258-B Natural Language Processing (6 CFU)
81678 Laboratory of Making (6 CFU)
96859 Embedded Systems and Internet of Things (6 CFU)
77780 Embedded Systems Programming and Internet of Things (6 CFU)
81683 Internet of Things (6 CFU)
90074 Smart Vehicular Systems (6 CFU)
93653 Iot And Big Data For Smart Agricultural Systems (5 CFU)
39176 Basic and Laboratory Informatics (3 CFU)
75198 Health Information Processing Systems (1 CFU)
20741 Information Processing Systems (1 CFU)
82202 Informatic (2 CFU)
82202-A Informatic (2 CFU)
93468 Distributed Software Systems (6 CFU)
87474 Distributed Systems (6 CFU)
91267 Multi-Agent Systems (6 CFU)
28024 Computer Networks T (9 CFU)
88325 Computer and Network Security Lab T (6 CFU)
B0833 Computer Security Laboratory T (6 CFU)
72942 Information Security M (8 CFU)
93466 FONDAMENTI DI CYBERSECURITY (6 CFU)
91254 Image Processing and Computer Vision (6 CFU)
73302 Computer Vision And Image Processing M (8 CFU)
93478 COMPUTER VISION (6 CFU)
69661 Image Processing And Computer Vision M (6 CFU)
B5722 Embedded Sensors (6 CFU)
96994 Smart Vehicular-Systems (6 CFU)
99195 Cyber-Physical Systems Programming M (6 CFU)
B0832 Administration of Systems T (6 CFU)
88324 System Administration T (6 CFU)
37760 Systems Simulation (6 CFU)
69731 Computer Architecture (12 CFU)
28011 Logic Design T (6 CFU)
11925 Computer Architecture (6 CFU)
95602 Programming and Computer Architectures (12 CFU)
28011-A Logic Design T (6 CFU)
96007 Foundations of Computer Architecture and Operating Systems (6 CFU)
88146 Fundamentals of Computer Programming P2 (3 CFU)
91685 History of Informatics and Computing Systems (6 CFU)
84401 Context-Aware Systems (6 CFU)
00819-B Programming (12 CFU)
00819 Programming (12 CFU)
00819-A Programming (9 CFU)
28623 Foundations of Informatics and Laboratory (12 CFU)
B1703 Introduction to Programming (12 CFU)
93034 Foundations of Computer Science T (12 CFU)
B5780 Computational Methods for Bioinformatics (10 CFU)
09730 Principles of Informatics (6 CFU)
29227 Foundations of Informatics T (9 CFU)
B8905 Foundations of Programming (6 CFU)
15305 Foundations of Informatics A (6 CFU)
27311-A Humanistic IT (6 CFU)
98735 Python Coding and Data Science (6 CFU)
89988 Elements of Informatics (6 CFU)
93034-A Foundations of Computer Science T (6 CFU)
16692 Laboratory of Informatics (3 CFU)
88145 Fundamentals of Computer Programming P1 (3 CFU)
B8159 INFORMATICA (8 CFU)
60348 Informatics (10 CFU)
90864 Elements of Computer Technologies for Fashion E-Commerce (6 CFU)
58423 Internet Architecture (9 CFU)
73273 Informatic Tecnologies for Fashion (6 CFU)
75969 Basic Informatics (6 CFU)
B8536 Computer-Aided Human Translation: CAT, MT and AI (5 CFU)
B8603 Machine Translation and Artificial Intelligence for Language Services (5 CFU)
73435 Project Management (6 CFU)
97431 Scalable and Reliable Services M (8 CFU)
B5796 Scalable and Reliable Services M (8 CFU)
09032 Software Engineering (6 CFU)
81612 Programming and Development paradigms (6 CFU)
B3567 INTRODUCTION TO QUANTUM COMPUTING (6 CFU)
28012-A Electronic Calculators T (6 CFU)
03716 Digital Design Principles and Computer Architecture (9 CFU)
28012 Electronic Calculators T (6 CFU)
91602 Computer Architectures (6 CFU)
95639 Industry 4.0 (6 CFU)
95632 Laboratory on Digital Transformation Technologies (3 CFU)
96725 IT and Game Localization (5 CFU)
B5474 ARCHITETTURE SOFTWARE A MICROSERVIZI (6 CFU)
72787 Mobile Application Programming (6 CFU)
66860 Mobile Applications Laboratory (6 CFU)
78779 Mobile Systems M (8 CFU)
95605 Web and Mobile Systems (6 CFU)
91250 DEEP LEARNING (6 CFU)
91250-A Deep Learning (6 CFU)
90477 Machine Learning Systems For Data Science (10 CFU)
85189 Systems and Algorithms for Data Science (10 CFU)
72796 Data Intensive Applications (6 CFU)
78778 Intelligent Systems M (8 CFU)
93319 INTRODUZIONE ALL'APPRENDIMENTO AUTOMATICO (6 CFU)
81610 Machine Learning (6 CFU)
88202 Machine Learning (6 CFU)
91250-B Deep Learning (6 CFU)
B1873 Machine Learning for Humanities (6 CFU)
96642 Virtualization and System Integration (6 CFU)
95646 Virtualized Systems (6 CFU)
91261 Artificial Intelligence in Industry (6 CFU)
B2125 Machine Learning and Data Mining M (8 CFU)
95631 Machine Learning and Data Mining (6 CFU)
28796 Elements of Databases (6 CFU)
30376 Business Intelligence (6 CFU)
B4940 DATA SCIENCE FOR FASHION SYSTEM (6 CFU)
96143 Methods and Models for Data Analytics M (6 CFU)
92858 Autonomous and Adaptive Systems M (8 CFU)
B0069 Artificial Intelligence for Digital Innovation (8 CFU)
81934 HUMAN COMPUTER INTERACTION (6 CFU)
B5059 Interaction Media Design (6 CFU)
93315 Computer Networks (12 CFU)
B0385 Natural Language Processing (6 CFU)
91258-A Natural Language Processing (6 CFU)
91258 Natural Language Processing (5 CFU)
B0073 ELABORAZIONE DI TESTI E GESTIONE DOCUMENTALE (8 CFU)
90749 Computing Education (6 CFU)
95965 Artificial Intelligence and Machine Learning for Business (6 CFU)
14015 Cryptography (6 CFU)
72778 High-Performance Computing (6 CFU)
58437 Parallel Calculus (6 CFU)
B8563 SISTEMI DI ELABORAZIONE ACCELERATA M (8 CFU)
B8562 SISTEMI CONCORRENTI E PARALLELI M (8 CFU)
90157 Open Science (6 CFU)
07276-A Computer Science (8 CFU)
85285 Programming (12 CFU)
65094 ADVANCED PROGRAMMING (6 CFU)
16870 Programming Laboratory (6 CFU)
41588 INTRODUZIONE ALL'ANALISI DEI DATI (6 CFU)
37262 Computer Programming (6 CFU)
28040 Information Technology (6 CFU)
B5727 Introduction to Languages for Artificial Intelligence (6 CFU)
B2696 Language, Technology, Research I: Computational Thinking (6 CFU)
93333 Programming Lab (3 CFU)
85446 Knowledge Representation and Extraction (6 CFU)
78096 Multimedia Data Management M (8 CFU)
91269 Multimedia Data Management (6 CFU)
77933 Multimedia Data Management M (6 CFU)
93469 Human Data Science (6 CFU)
95638 Operational Analytics (6 CFU)
88016 Statistics (2 CFU)
93470 Cybersecurity (6 CFU)
B5782 Algorithms and Data Structures for Computational Biology (10 CFU)
91411 Intelligent Robotic Systems (6 CFU)
81932 Big Data (6 CFU)
95630 Big Data and Cloud Platforms (6 CFU)
96142 Business Intelligence and Big Data M (6 CFU)
81942 Scalable and Cloud Programming (6 CFU)
91762 Combinatorial Decision Making and Optimization (6 CFU)
81609 Languages, Compilers and Computational Models (12 CFU)
66870 Concurrent Models and Systems (6 CFU)
17628 Concurrent and Distributed Programming (6 CFU)
84530 Database and Big Data Technologies M (8 CFU)
96346-A Informatics (6 CFU)
28021 Software Engineering T (9 CFU)
90106 Software Engineering (9 CFU)
B0061 Computer Methods for Digital Transformation (8 CFU)
66858 Software Engineering (6 CFU)
B1874 Project Management and Software Engineering for Cultural Heritage (6 CFU)
81672 Emerging Programming Paradigms (6 CFU)
91248 Fundamentals of Artificial Intelligence and Knowledge Representation (12 CFU)
98931 Fundamentals of Artificial Intelligence (6 CFU)
72939 Software Systems Engineering M (8 CFU)
17634 Computer Vision (6 CFU)
91266 Machine Learning for Computer Vision (6 CFU)
72945 Real-Time Systems M (8 CFU)
85665 Real Time Systems and Programming for Automation M (12 CFU)
78810 Real Time Systems for Automation M (12 CFU)
73025 Distributed Systems M (8 CFU)
91721 Laboratory of Virtual Reality and Advanced Reality (6 CFU)
90748 Blockchain and Cryptocurrencies (6 CFU)
91259 Architecture and Platforms for Artificial Intelligence (6 CFU)
11933 Theoretical Computer Science (6 CFU)
41169 Theoretical Computer Science (6 CFU)
08574-A Operating Systems (12 CFU)
28020 Operating Systems T (9 CFU)
88155 Industrial Informatics P (6 CFU)
58348 Lab-based Course on Operating Systems (6 CFU)
08574 Operating Systems (9 CFU)
85728 Real Time OS M (6 CFU)
76011 Multimedia Technologies in Archaeology (6 CFU)
81917 Interaction Multimedia Services (5 CFU)
84339 Advanced Database Systems (6 CFU)
98671 Business Analytics (3 CFU)
B8019 PROGRAMMING THE INTERNET OF THINGS M (8 CFU)
48693 Computer Science for Cultural Heritage (6 CFU)
B8561 ARCHITETTURE DEI CALCOLATORI M (8 CFU)
69430 COMPUTER ARCHITECTURES M (6 CFU)
95604 Operating Systems and Networks (6 CFU)
95644 Computer Networks and Network Programming (6 CFU)
41731-A Web-related Technologies (6 CFU)
85445 Information Modelling and Web Technologies (6 CFU)
95645 Foundations of Web Systems (6 CFU)
88566 Web Technology (9 CFU)
75835 Web Technologies (6 CFU)
28659 Web Technologies T (9 CFU)
41731 Web-related Technologies (6 CFU)
29227-B Foundations of Informatics T (6 CFU)
29227-A Foundations of Informatics T (6 CFU)
28027 Information Systems T (9 CFU)
96793 Information Systems and Database Management (6 CFU)
10906 Databases (6 CFU)
28652 IT Systems T-1 (9 CFU)
70155 Data Bases (9 CFU)
79194 Databases (6 CFU)
95611 Database Systems (6 CFU)
91161 Networks, Security And Data Bases (6 CFU)
B5793 Cybersecurity M (8 CFU)
93470-A Cybersecurity (6 CFU)
93669 Intelligent Systems Engineering (6 CFU)
90147 Artificial intelligence (6 CFU)
97466 Digital represention of the Landscape Project (6 CFU)
96995 Programmable Networking (6 CFU)
37085 Principles, Models and Applications for Distributed Systems M (6 CFU)
84531 Infrastructures For Cloud Computing and Big Data M (8 CFU)
10906-A Databases (12 CFU)
90107 BASI DI DATI (9 CFU)
B4913 ALGORITMI E STRUTTURE DATI (9 CFU)
09730-A Principles of Informatics (6 CFU)
B0070 METODOLOGIE E TECNICHE DI SIMULAZIONE (8 CFU)
42500 Semantic Web (6 CFU)
91265 Knowledge Engineering (6 CFU)
B8929 Computer Studies and Human Sciences (6 CFU)
B5292 Knowledge Engeneering for the Humanities (3 CFU)
B5293 Computer Studies and Humanities (3 CFU)
B0063 Programming Laboratory (4 CFU)
11929 Algorithms and Data Structures (12 CFU)
11929-A Algorithms and Data Structures (12 CFU)
37635 Data Structure and Algorithms (12 CFU)
11929-B Algorithm and Data Structures (6 CFU)
85301 Algorithms and Data Structures in Biology (6 CFU)
28004 Foundations of Informatics T-1 (9 CFU)
96346 INFORMATICA (6 CFU)
70226 Network Programming (6 CFU)
B4948 Cognitive Sematics for Artificial Intelligence (6 CFU)
B8836 Universal Semantics (6 CFU)
93379 Digital Culture (9 CFU)
B8105 Computational Management of Data (12 CFU)
94442 Design of Web Applications T (6 CFU)
48693-A Computer Science for Cultural Heritage (6 CFU)
B5079 Digital Humanities and Data Management for Cultural Heritage (6 CFU)
73149 Logic for Informatics (6 CFU)
90720 Usability & User Experience Design (6 CFU)
91264 User Experience Design (6 CFU)
85573 Usability and User Experience (6 CFU)
04138 Programming Languages (12 CFU)
72940 Computational Models And Languages M (8 CFU)
72671 Complements of Programming Languages (6 CFU)
B0064 Introduction to Data Science and Computational Thinking (8 CFU)
87469 Decision Making with Constraint Programming (6 CFU)
72938 Foundations Of Artificial Intelligence T (8 CFU)
81940 Artificial Intelligence (6 CFU)
93467 Cryptography (6 CFU)
B5812 Cybersecurity Governance (8 CFU)
98618 "Document Digitization Techniques " (6 CFU)
B5674 Management of Digital Media (5 CFU)
70219 Object-Oriented Programming (12 CFU)
B8554 SISTEMI SOFTWARE A OGGETTI T (12 CFU)
95648 Software Design and Development (9 CFU)
08574-B Operating Systems (12 CFU)
90730 Social Network Analysis (6 CFU)
95782 Network Analysis (6 CFU)
37925 Virtual System Design (6 CFU)
69867 Web Services and Applications (6 CFU)
27311 Humanistic IT (6 CFU)
66857 Logic for Computer Science (6 CFU)
09679 Data Base Sistems (6 CFU)
B0074 PIATTAFORME PER LA TRASFORMAZIONE DIGITALE (8 CFU)
B1459 ARTIFICIAL INTELLIGENCE, BLOCKCHAIN E CRIPTOVALUTE NELLO SVILUPPO SOFTWARE (6 CFU)
69259 Computer Science (6 CFU)
87898 Methods and Technologies for Interpreting (5 CFU)
B4925 IMMERSIVE TECHNOLOGIES FOR FASHION (6 CFU)
B6082 Technology-Assisted Interpreting I (8 CFU)
87901 Methods and Advanced Technologies for Interpreting (5 CFU)
B5026 Introduction to Computability and Complexity (6 CFU)
47240 Web Systems Engineering (6 CFU)
1: effectful_programming, monadic_programming, acceptance_test_driven_development, ... (20 topics, est. 5.4 ECTS; 11.0 current CFU)
2: activity_aware_systems, ubiquitous_systems, behavior_change_models, ... (19 topics, est. 4.7 ECTS; 7.0 current CFU)
3: chain_of_custody, cloud_forensics, command_line_systems, ... (18 topics, est. 5.7 ECTS; 6.0 current CFU)
4: productivity_software, computer_software_basics, operating_systems, ... (17 topics, est. 3.7 ECTS; 31.4 current CFU)
5: domain_driven_design, collaborative_software_development, software_license, ... (16 topics, est. 5.7 ECTS; 15.1 current CFU)
6: large_language_models, language_models, prompt_engineering, ... (16 topics, est. 4.5 ECTS; 28.5 current CFU)
7: computational_linguistics, n_gram_language_models, speech_processing, ... (15 topics, est. 3.6 ECTS; 4.5 current CFU)
8: embedded_systems, iot_systems, wireless_communication_systems, ... (15 topics, est. 5.7 ECTS; 24.3 current CFU)
9: healthcare_information_systems, healthcare_data_archiving, digital_document_lifecycle, ... (15 topics, est. 2.2 ECTS; 5.8 current CFU)
10: cap_theorem, distributed_consistency, distributed_middleware, ... (14 topics, est. 5.0 ECTS; 20.3 current CFU)
11: cryptography, authentication_models, network_firewalls, ... (14 topics, est. 5.7 ECTS; 30.5 current CFU)
12: image_segmentation, image_formation_acquisition, image_convolution, ... (14 topics, est. 4.9 ECTS; 22.4 current CFU)
13: analog_digital_conversion, dimensionality_reduction, information_theory, ... (13 topics, est. 4.7 ECTS; 9.6 current CFU)
14: autonomous_systems, robotics, cyber_physical_systems, ... (13 topics, est. 4.8 ECTS; 10.8 current CFU)
15: centralized_monitoring, network_configuration_management, network_configuration_protocols, ... (13 topics, est. 4.2 ECTS; 17.9 current CFU)
16: agent_based_simulation, complex_systems_analysis, discrete_event_simulation, ... (12 topics, est. 5.5 ECTS; 7.8 current CFU)
17: computer_hardware_architecture, binary_encoding, digital_logic_networks, ... (12 topics, est. 5.8 ECTS; 64.0 current CFU)
18: edge_fog_computing, mobile_middleware, positioning_systems, ... (12 topics, est. 3.4 ECTS; 5.2 current CFU)
19: imperative_constructs, arrays_matrices, functions_recursion, ... (12 topics, est. 5.9 ECTS; 99.7 current CFU)
20: internet_world_wide_web, computer_networks, it_security, ... (12 topics, est. 5.8 ECTS; 35.9 current CFU)
21: machine_translation, project_management, translation_post_editing, ... (12 topics, est. 6.0 ECTS; 10.9 current CFU)
22: software_testing, software_lifecycle, software_quality_assurance, ... (12 topics, est. 5.8 ECTS; 29.5 current CFU)
23: complex_hilbert_spaces, linear_algebra, quantum_algorithms, ... (11 topics, est. 5.5 ECTS; 5.9 current CFU)
24: cpu_organization, risc_cisc_architectures, memory_io_addressing, ... (11 topics, est. 5.2 ECTS; 29.4 current CFU)
25: digital_transformation_design, technology_assessment, digitalization_impacts, ... (11 topics, est. 4.0 ECTS; 7.4 current CFU)
26: localization_accessibility, localization_bug_prevention, localization_editing_revision, ... (11 topics, est. 5.0 ECTS; 5.0 current CFU)
27: microservices_architecture, docker_containerization, service_oriented_computing, ... (11 topics, est. 3.5 ECTS; 11.2 current CFU)
28: mobile_application_development, mobile_device_features, mobile_networking_services, ... (11 topics, est. 5.5 ECTS; 13.8 current CFU)
29: neural_networks_deep_learning, machine_learning, convolutional_neural_networks, ... (11 topics, est. 6.0 ECTS; 63.0 current CFU)
30: virtual_machines, virtual_networks, distributed_file_systems, ... (11 topics, est. 3.4 ECTS; 11.9 current CFU)
31: anomaly_detection, epidemic_control, fairness_constraints, ... (10 topics, est. 4.0 ECTS; 5.4 current CFU)
32: classification_algorithms, clustering, data_mining, ... (10 topics, est. 4.8 ECTS; 34.4 current CFU)
33: generative_models, reinforcement_learning, ai_ethics, ... (10 topics, est. 5.5 ECTS; 21.9 current CFU)
34: interaction_models, mental_models, prototyping, ... (10 topics, est. 4.0 ECTS; 7.2 current CFU)
35: link_layer_protocols, wireless_networking, mobility_management, ... (10 topics, est. 3.4 ECTS; 9.1 current CFU)
36: text_classification, natural_language_processing, word_embeddings, ... (10 topics, est. 3.7 ECTS; 16.4 current CFU)
37: computer_science_education, cs_curriculum_design, cs_education_pedagogy, ... (9 topics, est. 5.3 ECTS; 5.9 current CFU)
38: data_analytics, sustainability_esg, automated_decision_systems, ... (9 topics, est. 3.4 ECTS; 6.7 current CFU)
39: digital_signatures, digital_certificates, cryptographic_key_management, ... (9 topics, est. 5.5 ECTS; 11.6 current CFU)
40: hpc_architectures, shared_memory_programming, gpu_programming, ... (9 topics, est. 5.1 ECTS; 17.4 current CFU)
41: information_retrieval, similarity_measures, data_preprocessing, ... (9 topics, est. 2.3 ECTS; 15.3 current CFU)
42: instance_detection, transfer_learning, connected_components_blob_analysis, ... (9 topics, est. 3.0 ECTS; 7.9 current CFU)
43: open_access, open_data, open_infrastructures, ... (9 topics, est. 4.5 ECTS; 4.5 current CFU)
44: programming_in_python, basic_data_types, functions, ... (9 topics, est. 4.0 ECTS; 65.5 current CFU)
45: semantic_inference, knowledge_extraction, ontology_design_patterns, ... (9 topics, est. 4.0 ECTS; 8.3 current CFU)
46: social_networks, user_interface_design, multimedia_data_management, ... (9 topics, est. 4.2 ECTS; 25.2 current CFU)
47: statistical_data_analysis, predictive_analytics, data_pipeline_design, ... (9 topics, est. 3.3 ECTS; 15.8 current CFU)
48: application_security, device_hardening, network_security, ... (8 topics, est. 2.7 ECTS; 7.6 current CFU)
49: approximation_algorithms, branch_and_bound, burrows_wheeler_transform, ... (8 topics, est. 4.1 ECTS; 4.6 current CFU)
50: associative_learning, behavior_based_robotics, evolutionary_robotics, ... (8 topics, est. 3.7 ECTS; 3.7 current CFU)
51: big_data_management, nosql_databases, spark_computing, ... (8 topics, est. 4.3 ECTS; 22.2 current CFU)
52: combinatorial_optimization, constraint_programming, constraint_propagation, ... (8 topics, est. 5.2 ECTS; 11.6 current CFU)
53: concurrency_formal_modeling, labeled_transition_systems, bisimulation_equivalence, ... (8 topics, est. 5.6 ECTS; 7.7 current CFU)
54: data_models, nosql_data_model_design, database_resource_optimization, ... (8 topics, est. 3.8 ECTS; 12.7 current CFU)
55: dematerialization, digital_transformation_regulation, documentary_databases, ... (8 topics, est. 5.8 ECTS; 6.9 current CFU)
56: design_patterns, software_engineering, requirements_analysis, ... (8 topics, est. 4.7 ECTS; 41.5 current CFU)
57: functional_programming, type_checking, certified_programming, ... (8 topics, est. 3.7 ECTS; 10.2 current CFU)
58: knowledge_based_systems, production_rule_systems, bayesian_networks, ... (8 topics, est. 4.4 ECTS; 11.0 current CFU)
59: kotlin_programming, model_driven_software_development, automatic_code_generation, ... (8 topics, est. 3.0 ECTS; 9.5 current CFU)
60: object_detection, metric_learning, vision_transformers, ... (8 topics, est. 4.2 ECTS; 7.1 current CFU)
61: real_time_systems, real_time_scheduling, real_time_task_models, ... (8 topics, est. 5.9 ECTS; 22.7 current CFU)
62: asynchronous_programming, dependency_injection_aop, distributed_clustering, ... (7 topics, est. 3.1 ECTS; 4.8 current CFU)
63: avatar_systems, computer_graphics, distributed_virtual_environments, ... (7 topics, est. 2.7 ECTS; 3.1 current CFU)
64: blockchain_mining, blockchain_privacy, blockchain_security, ... (7 topics, est. 3.8 ECTS; 3.8 current CFU)
65: cloud_ml_platforms, heterogeneous_ml_architectures, ml_algorithmic_optimization, ... (7 topics, est. 2.1 ECTS; 2.1 current CFU)
66: computability_limits, abstract_data_types, turing_machines, ... (7 topics, est. 3.6 ECTS; 29.2 current CFU)
67: concurrent_computing, process_synchronization, cpu_scheduling, ... (7 topics, est. 4.5 ECTS; 45.8 current CFU)
68: cultural_heritage_communication, three_dimensional_modeling, photogrammetry, ... (7 topics, est. 3.7 ECTS; 6.4 current CFU)
69: data_backup, design_thinking, intellectual_property, ... (7 topics, est. 1.6 ECTS; 2.2 current CFU)
70: database_administration, active_databases, business_intelligence, ... (7 topics, est. 3.4 ECTS; 7.1 current CFU)
71: device_description_languages, intelligent_system_perception_actuation, iot_platforms, ... (7 topics, est. 4.4 ECTS; 5.5 current CFU)
72: digital_images, virtual_tours, gis, ... (7 topics, est. 3.6 ECTS; 5.6 current CFU)
73: hardware_virtualization, memory_hierarchy, shared_memory_architectures, ... (7 topics, est. 4.2 ECTS; 12.4 current CFU)
74: internet_routing, internet_architecture, network_quality_of_service, ... (7 topics, est. 2.8 ECTS; 8.6 current CFU)
75: markup_languages, http_web_protocols, json_data_format, ... (7 topics, est. 4.4 ECTS; 33.2 current CFU)
76: programming_in_cpp, numerical_analysis, programming_functions, ... (7 topics, est. 2.9 ECTS; 13.2 current CFU)
77: relational_data_model, relational_database_design, databases_sql, ... (7 topics, est. 3.9 ECTS; 40.8 current CFU)
78: risk_analysis, security_auditing, security_policies, ... (7 topics, est. 4.0 ECTS; 14.5 current CFU)
79: artificial_intelligence, knowledge_representation, ai_planning, ... (6 topics, est. 2.6 ECTS; 25.2 current CFU)
80: cad_technical_drawing, raster_graphics, vector_graphics, ... (6 topics, est. 3.8 ECTS; 5.7 current CFU)
81: cloud_computing, distributed_scalability, cloud_resource_federation, ... (6 topics, est. 3.6 ECTS; 15.8 current CFU)
82: cloud_service_models, stream_processing, data_centers, ... (6 topics, est. 2.2 ECTS; 5.9 current CFU)
83: database_transactions, relational_algebra, database_architectures, ... (6 topics, est. 4.1 ECTS; 26.3 current CFU)
84: hash_tables, heaps_priority_queues, recurrence_equations, ... (6 topics, est. 3.8 ECTS; 9.9 current CFU)
85: numeric_information_representation, von_neumann_architecture, non_numeric_information_representation, ... (6 topics, est. 2.4 ECTS; 10.3 current CFU)
86: ontology_engineering, semantic_web, knowledge_graphs, ... (6 topics, est. 3.3 ECTS; 24.9 current CFU)
87: python_data_science_libraries, regression_analysis, data_visualization, ... (6 topics, est. 2.4 ECTS; 24.4 current CFU)
88: algorithm_design_analysis, data_structures, sorting_algorithms, ... (5 topics, est. 2.7 ECTS; 75.5 current CFU)
89: client_server_model, tcp_ip_protocols, transport_layer_protocols, ... (5 topics, est. 2.1 ECTS; 14.4 current CFU)
90: cognitive_science, multimodal_data_interpretation, semiotics, ... (5 topics, est. 4.8 ECTS; 9.3 current CFU)
91: decision_quality_monitoring, digital_data_acquisition_storage, digital_decision_making, ... (5 topics, est. 2.9 ECTS; 5.0 current CFU)
92: dynamic_programming, greedy_algorithms, brute_force_algorithms, ... (5 topics, est. 2.4 ECTS; 9.0 current CFU)
93: enterprise_components, enterprise_persistence, software_architecture, ... (5 topics, est. 2.9 ECTS; 12.9 current CFU)
94: fair_data_management, digital_humanities, digital_resource_design, ... (5 topics, est. 5.2 ECTS; 12.4 current CFU)
95: first_order_logic, propositional_logic, action_logic, ... (5 topics, est. 3.7 ECTS; 8.6 current CFU)
96: human_computer_interaction, interaction_design, usability_evaluation, ... (5 topics, est. 3.5 ECTS; 24.2 current CFU)
97: language_implementation, lexical_analysis, context_free_grammars_parsing, ... (5 topics, est. 2.9 ECTS; 13.4 current CFU)
98: object_mutability, scope_visibility, mathematical_logic, ... (5 topics, est. 2.4 ECTS; 9.2 current CFU)
99: programming_languages, computational_thinking, regular_expressions, ... (5 topics, est. 2.6 ECTS; 12.4 current CFU)
100: search_algorithms, constraint_satisfaction, logic_programming, ... (5 topics, est. 3.7 ECTS; 22.2 current CFU)
101: asymptotic_analysis, graph_algorithms, programming_in_java, ... (4 topics, est. 3.7 ECTS; 21.4 current CFU)
102: biomolecular_sequence_structure_analysis, genomic_annotation, heuristic_alignment, ... (4 topics, est. 2.5 ECTS; 2.5 current CFU)
103: cryptographic_protocol_verification, perfect_security, pseudorandomness, ... (4 topics, est. 4.0 ECTS; 4.0 current CFU)
104: cyber_risk_management, cybersecurity_governance, disinformation, ... (4 topics, est. 4.0 ECTS; 4.0 current CFU)
105: digital_image_quality_assessment, document_digitization, scanner_calibration, ... (4 topics, est. 2.7 ECTS; 2.7 current CFU)
106: digital_media_economics, digital_media_transformation, media_market_dynamics, ... (4 topics, est. 5.0 ECTS; 5.0 current CFU)
107: exceptions, graphical_user_interfaces, java_language_features, ... (4 topics, est. 2.5 ECTS; 9.2 current CFU)
108: file_systems, shell_scripting, os_security, ... (4 topics, est. 3.0 ECTS; 16.2 current CFU)
109: information_system_classification, information_systems, data_flow_diagrams, ... (4 topics, est. 2.2 ECTS; 5.9 current CFU)
110: network_analysis, network_data_management, network_research_design, ... (4 topics, est. 6.0 ECTS; 12.0 current CFU)
111: parallel_architectures, concurrent_language_programming, hardware_multithreading, ... (4 topics, est. 2.2 ECTS; 5.4 current CFU)
112: user_level_operating_systems, virtual_devices, virtual_file_systems, ... (4 topics, est. 3.4 ECTS; 3.4 current CFU)
113: web_architectures_solution_stacks, web_language_supersets, asynchronous_web_notifications, ... (4 topics, est. 2.7 ECTS; 4.1 current CFU)
114: web_distribution_frameworks, digital_content_production, ict_project_management, ... (4 topics, est. 3.5 ECTS; 7.1 current CFU)
115: agile_software_development, continuous_integration, software_build_automation (3 topics, est. 1.5 ECTS; 11.2 current CFU)
116: ai_cybersecurity, cyber_conflict, defensive_security (3 topics, est. 2.9 ECTS; 3.8 current CFU)
117: autoencoders, automatic_differentiation, calculus (3 topics, est. 1.3 ECTS; 1.3 current CFU)
118: axiomatic_set_theory, formal_logic_foundations, structural_induction_recursion (3 topics, est. 3.0 ECTS; 3.0 current CFU)
119: business_process_reengineering, cost_benefit_analysis, database_integration_master_data (3 topics, est. 2.2 ECTS; 2.2 current CFU)
120: cognitive_models, cryptocurrencies, software_quality_models (3 topics, est. 2.5 ECTS; 5.8 current CFU)
121: conceptual_blending, semantic_universals, theory_of_mind (3 topics, est. 3.0 ECTS; 3.0 current CFU)
122: data_collection_analysis, numerical_cartography, server_side_application_development (3 topics, est. 2.0 ECTS; 2.0 current CFU)
123: documentation_strategies, specialized_corpora, terminology_extraction (3 topics, est. 3.0 ECTS; 3.0 current CFU)
124: exhaustive_search, motif_finding, restriction_mapping (3 topics, est. 1.6 ECTS; 1.6 current CFU)
125: fault_tolerance_reliability, web_api_design, data_serialization (3 topics, est. 1.0 ECTS; 6.3 current CFU)
126: immersive_platforms, immersive_technologies, virtual_reality (3 topics, est. 2.8 ECTS; 4.8 current CFU)
127: language_syntax, language_semantics, language_evolution (3 topics, est. 1.6 ECTS; 9.1 current CFU)
128: live_subtitling, speech_recognition, terminology_management (3 topics, est. 5.5 ECTS; 14.2 current CFU)
129: pac_learning, uniform_convergence, vc_dimension (3 topics, est. 3.0 ECTS; 3.0 current CFU)
130: web_client_side_programming, web_server_side_programming, web_frameworks_content_management (3 topics, est. 2.1 ECTS; 16.3 current CFU)
131: code_transformation, symbol_tables (2 topics, est. 1.7 ECTS; 1.7 current CFU)
132: distributed_communication, distributed_systems (2 topics, est. 0.7 ECTS; 15.0 current CFU)
133: r_functions, r_graphics_packages (2 topics, est. 0.8 ECTS; 0.8 current CFU)
134: web_build_tools, web_mobile_application_types (2 topics, est. 1.6 ECTS; 2.4 current CFU)